Assessment of the SeF_6 ion-TPC proposal: noise, collection time, pile-up and calorimetry

NAUSICAA working note

4 November 2026

1 Assessment and scope

Nygren et al. [1] establish an attractive physics case: the high ^{82}Se Q value suppresses common gamma backgrounds, thermal ions diffuse little, and direct collection avoids multiplication noise. The difficult part is simultaneously obtaining fine tracking, complete charge collection and low-noise calorimetry for slow, distributed charge. The original sub-50-electron ENC is an engineering projection. Later Topmetal-S measurements in the local literature are about 115–140 electrons, and the NνDEx conceptual design explicitly identifies two-neutrino coincidence backgrounds for enriched gas.

This note distinguishes measured electronics performance, design assumptions, and analytical estimates. It is not a new SeF_6 transport or track simulation. The calculations use $Q = 2.995$ MeV, $W = 32$ eV and $F = 0.19$ from the original paper; W and F were inferred using SF_6, rather than measured in the proposed high-pressure SeF_6 detector. Track extents below are illustrative.

2 What supports the CSA noise assumption?

Section 4 and figure 7 of Nygren et al. propose a collection electrode smaller than approximately 1 mm, a few pF of input capacitance, a femtofarad feedback capacitor and an almost-off feedback transistor with adjustable resistance. Charge focusing allows a small collection electrode at a several-mm to cm pixel pitch. Direct waveform digitization replaces a long analog shaper; timing comes from the rising edge and energy from digital filtering. They require a long signal retention time and good noise at that integration time. They do not give a measured ENC-versus-shaping-time curve for the proposed few-pF electrode, a complete parallel/series noise budget, or a calibrated MeV calorimetric demonstration at 10 bar.

Their cited Topmetal-II$^-$ demonstration [2] reported 13.9 electrons ENC with a 1 ms digital trapezoidal filter and 7.6 ms feedback decay. This used a 25×25 μm collection node and approximately 23 fF input capacitance. The injection capacitance of 5.5 fF came from simulation and was explicitly not independently calibrated. This is evidence for direct ion sensing, but it does not establish the same ENC for a roughly 3 pF electrode and a much longer input pulse.

The later local papers provide more concrete information:

- Yang et al. [3], sections 2–3, describe a 130 nm CMOS folded-cascode CSA, PMOS input transistor chosen for flicker noise, adjustable feedback and external digitization. They measure input capacitance near 3 pF, injection capacitance near 250 fF, gain around 220 mV/fC and ENC around 115 electrons after digital trapezoidal shaping. They explicitly state that their required ENC below 45 electrons for the target resolution has not yet been met.

- Liang et al. [4], section 2.2, measure 121 ± 9 electrons ENC after calibrating the injection capacitance against external capacitors. They report 255 ± 10 fF and explain that earlier underestimation of this capacitance led to underestimated noise in earlier Topmetal-S reports. This finding should not be applied as an unmeasured numerical correction to Topmetal-II$^-$,

which is a different device. ENC rises to about 140 electrons as feedback decay is shortened from 200 to 10 ms; the pulse rise time is about 3 ms. The tested feedback decay range extends to about 1.5 s. A long decay time alone does not establish low ENC for every slow charge waveform. Ion demonstrations use air and SF_6 near atmospheric pressure, not a complete 10-bar SeF_6 calorimeter.

- The 2025 low-leakage study [5], sections III–IV, demonstrates charge retention of tens of seconds and sub-0.6 fA inferred leakage. Its low-noise CSA plus switch performance is a separate simulation: 29 electrons for the CSA, 487 with the switch, and 39 after an ideal noiseless CR–RC filter. The simulated filtered ENC grows to 52 electrons as the injection duration reaches 300 ms. Thus 39 electrons is not a measured system ENC for second-long high-pressure events.

The physical difficulty is the trade-off between bandwidth, input capacitance, leakage/feedback noise, flicker noise and baseline stability. Suppressing white series noise by longer filtering does not also suppress all parallel or low-frequency noise. As an illustrative charge-counting limit only, independent Poisson leakage contributes

$$\sigma_{\mathrm{leak}}^2 = I_{\mathrm{leak}} T/e. \tag{1}$$

For a constant 0.6 fA leakage current integrated for 1 s this gives 61.2 electrons RMS. This is not a prediction for the switch circuit: its feedback correlations and actual filter must be included.

3 Track length and drift times

Nygren section 5 compares 10-bar gases at their respective endpoints: about 2.99 MeV for ^{82}Se and 2.46 MeV for ^{136}Xe. The paper argues for less multiple scattering in the lower-Z mixture and longer tracks at the higher endpoint. It does not provide a quantitative same-energy range comparison or distributions of track length and Δz for both gases. Figure 8 illustrates centimetre-scale extents; the CDR uses approximately 10 cm as a characteristic $0\nu\beta\beta$ event size at 10 bar.

Consequently, the qualitative topology advantage is supported, but a generic claim that SeF_6 has a longer electron stopping range at equal pressure and equal energy is not established by these papers. Equal pressure fixes molecule/atom number density at a fixed temperature, not mass density or stopping power. Less multiple scattering, true path length, endpoint separation and projection along z are distinct quantities. A quantitative gas comparison would require matched transport simulations and orientation distributions. One displayed event is not an ensemble range measurement.

For an ion mobility K and field E, distinguish full drift latency from the arrival spread of one species across a track:

$$v = KE, \qquad t_{\mathrm{drift}} = L/v, \qquad \Delta t_{\mathrm{track}} = \Delta z/v. \tag{2}$$

For reduced mobility K_0 referenced to (P_0, T_0),

$$K = K_0 \frac{P_0}{P} \frac{T}{T_0}. \tag{3}$$

Using Nygren's approximate $K_0 = 0.4$–0.6 cm^2/(V s), 10 bar, 293.15 K and 400 V/cm gives 0.174–0.261 m/s. This extrapolates SF_6 information to SeF_6. The CDR adopts a speed above 20 cm/s and approximately 7 s full drift over 1.6 m [6]. The original paper's 0.5–1 s ton-scale drift estimate and mm/ms language should not be used without this density correction; they are optimistic compared with its own reduced mobilities at 10 bar and the later CDR.

Extent Δz (mm)	$v = 0.1$ m/s: time (s)	$v = 0.2$ m/s: time (s)
10	0.1	0.05
20	0.2	0.1
100	1	0.5
200	2	1
1600	16	8

The 0.1 m/s column is the current NAUSICAA adopted Xe-ion speed, a design assumption rather than a measured equality of Xe and SeF_6 mobilities. Thus 10–20 mm implies 0.05–0.2 s at these representative speeds. A 1–2 s interval is appropriate for 10–20 cm at 0.1 m/s, or requires a slower speed of 1 cm/s for the original 10–20 mm extent. Diffusion, multiple ion species and electronics response can broaden the observed waveform beyond the single-species geometrical estimate.

An additional primary-source check is the 2026 simulation framework [7], which calculates reduced SeF_5^- and SeF_6^- mobilities of 0.444 and 0.430 $cm^2/(V\ s)$. These are calculations, not high-pressure SeF_6 measurements. It explicitly treats the two species as a hypothesis because their yields and the lifetime of SeF_6^- in pure high-pressure gas remain unverified.

4 Pile-up and the absence of prompt timing

The relevant event rate includes the entire low-energy background and two-neutrino spectrum, not just the very small single-event rate at Q. For a local Poisson rate r and unresolved time interval T, the probability of another event is $1 - e^{-rT}$, approximately rT at low occupancy. The ambiguous spacetime volume also matters: simultaneous events on distant channels need not be confused. Full drift latency does not equal mandatory global acquisition dead time in a continuously digitized TPC.

The CDR section IV.E explicitly estimates the coincidence problem. Assuming events are separable when approximately $10 \times 10 \times 10$ cm^3 apart, its approximate $2\nu\beta\beta + 2\nu\beta\beta$ background in the ROI is 0.06 events/year for 100 kg natural SeF_6 and about 8 events/year for enriched gas. The latter exceeds its approximately 0.4 events/year single-event gamma estimate before topological rejection. These are CDR projections under their energy-window and separation assumptions, not new rates inferred from our timing table. Their proposed remedies include topology studies and potentially prompt scintillation readout, whose light yield and practical implementation require study.

Two distinct negative-ion mobilities can in principle provide an absolute drift distance from

$$\Delta t_{12} = L(1/v_1 - 1/v_2). \tag{4}$$

This helps fiducialization without a prompt t_0, but requires identifying and pairing both species. It does not automatically associate all charge unambiguously when two extended tracks overlap. The 2026 framework retains this dual-species assumption explicitly. A complete pile-up assessment must include the waveform, threshold, local occupancy and pairing ambiguity.

5 Energy-resolution budget

For $N = Q/W = 93594$ primary charges and one complete-event charge estimate per selected channel, assume independent identical RMS noise ENC. Adding independent variance gives

$$R_{\mathrm{FWHM}} = 2\sqrt{2\ln 2}\,\frac{\sqrt{FN + n\,\mathrm{ENC}^2}}{N}. \tag{5}$$

This is an optimistic instrumental budget: it assumes complete collection, perfect gain/baseline calibration, no thresholds, no recombination or attachment fluctuations, no pile-up and independent channel estimates. The intrinsic FWHM is 0.336%.

Independent channels n	30 electrons ENC	50 electrons ENC	120 electrons ENC
25	0.505%	0.713%	1.546%
100	0.826%	1.302%	3.038%
300	1.350%	2.205%	5.240%
400	1.546%	2.538%	6.048%

The CDR's 25 channels at 30 electrons reproduces its quoted 0.51% FWHM. Nygren's few-hundred-pixel assumption yields a considerably less favorable budget even at 50 electrons. In particular, a total electronic noise near 1000 electrons gives 2.54% FWHM including Fano noise; being below 1000 electrons is not by itself sufficient to ensure below 2%. The maximum independent-channel budgets before additional imperfections are:

Target FWHM	Total noise budget (electrons)	Max. n, ENC 50	Max. n, ENC 120
0.5%	147.3	8.7	1.5
0.7%	244.2	23.8	4.1
1.0%	374.4	56.1	9.7
2.0%	783.6	245.6	42.6

These are continuous variance budgets; an integer channel limit is their floor.

For correlated channel/time estimates, replace $n\,\mathrm{ENC}^2$ by $\boldsymbol{w}^T C \boldsymbol{w}$, using the actual estimator weights and noise covariance. Sample counts cannot be equated to independent noise measurements. Conversely, a short injected-pulse ENC cannot simply be used once per pixel for an extended, time-dependent ion current without verifying the complete estimator. Threshold selection can discard low-density track charge and produce energy bias and non-Gaussian tails; the original variance budget does not cover these effects. Measuring both polarities can reduce independent instrumental noise but not the shared Fano fluctuation.

6 Implications and reproducibility

The strongest demonstrated advantages are high endpoint energy and thermal ion diffusion. The limiting engineering evidence concerns noise at realistic input capacitance and waveform duration, reliable collection and charge reconstruction, and enriched-gas pile-up. A convincing calorimetric validation should inject known charge with realistic time distributions, measure the full channel covariance and baseline stability, and report total reconstructed energy as channel multiplicity and duration vary. A matched SeF_6/Xe track study would resolve the separate question of z-extent distributions.

Numerical inputs are in `config/nygren_review.toml`. Julia saves SI-based calculations, source hashes and code in `results/nygren_review/budgets.h5`; Python presents the saved values. Reproduce from the repository root:

```
JULIA_DEPOT_PATH="$PWD/work/julia-depot:$HOME/.julia" \
  julia --startup-file=no --project=. scripts/validation/nygren_review.jl
python3.13 scripts/validation/present_nygren_review.py
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -outdir=latex_generated/nygren_review docs/nygren_ion_tpc_review.tex
```

References

[1] D.R. Nygren, B.J.P. Jones, N. López-March, Y. Mei, F. Psihas, and J. Renner. Neutrinoless double beta decay with $^{82}SeF_6$ and direct ion imaging. *Journal of Instrumentation*, 13(03): P03015, March 2018. ISSN 1748-0221. doi: 10.1088/1748-0221/13/03/p03015. URL `http://dx.doi.org/10.1088/1748-0221/13/03/P03015`.

[2] M. An, C. Chen, C. Gao, M. Han, R. Ji, X. Li, et al. A low-noise cmos pixel direct charge sensor, topmetal-ii-. *Nuclear Instruments and Methods in Physics Research Section A*, 810: 144–150, 2016.

[3] Y. Yang, T. Liang, C. Gao, et al. Design and preliminary test results of the charge sensitive amplifier for gain-less charge readout in high-pressure TPC. *Journal of Instrumentation*, 19: C03031, 2024. doi: 10.1088/1748-0221/19/03/C03031.

[4] Tianyu Liang, Dongliang Zhang, Hulin Wang, Chaosong Gao, Jun Liu, Xiangming Sun, Le Xiao, Feng Liu, Bihui You, Ling Liu, Yichen Yang, and Kai Chen. Performance of a novel charge sensor on the ion detection for the development of a high-pressure avalancheless ion TPC. *Journal of Instrumentation*, 19(04):C04004, 2024. doi: 10.1088/1748-0221/19/04/C04004.

[5] Xu Wang, Yujie Li, Tong Wan, et al. Low leakage switch technology for ion signal detection for NνDEx. ChinaXiv:202509.00114v1, 2025. URL `https://chinaxiv.org/abs/202509.00114`. Preprint posted 9 September 2025.

[6] Xi-Guang Cao et al. NνDEx-100 conceptual design report. *Nucl. Sci. Tech.*, 35(1):3, 2024. doi: 10.1007/s41365-023-01360-7.

[7] Tianyu Liang, Hulin Wang, Dongliang Zhang, Chaosong Gao, Xiangming Sun, Feng Liu, Jun Liu, Chengui Lu, Yichen Yang, Chengxin Zhao, Hao Qiu, and Kai Chen. Development of a simulation and analysis framework for NνDEx experiment. *Nuclear Instruments and Methods in Physics Research Section A*, 1087:171439, 2026. doi: 10.1016/j.nima.2026.171439. URL `https://arxiv.org/abs/2508.13582v4`.